

Gesture as a Brush

Translating Hand Motion into
AI-Generated Paintings

CAS AICP Module 5 Exercise
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Idea:

Translate hand motion into abstract art using cellphones and diffusion models

Exercise assumptions and questions

- (1) Motion is fundamental to human expression and hence an engine for many art forms.
- (2) AI can generate images, but can it interpret gestures (brush strokes)?
- (3) Can embodied movement become a controllable artistic input?
- (4) Can I create my own application for this by «vibe-coding» in Python using GPT5?

Key milestones

- Sensor recording (phyphox)
- Feature extraction
- Machine learning - training (scikit-learn)
- Prediction - test set
- Selection of correctly predicted paintings
- Prompt creation based on features
- Image generation using diffusion model (Pytorch, Stable Diffusion v1.5)

Vibecoding: *Refining the project idea*

Adjustments to the initially generated code (examples)

- Each imported Phyphox dataset should represent one individual painting.
- Export a plot that shows a “motion fingerprint” for each imported painting.
- Can the program later on also guess what the user was trying to paint?
- Generate only correctly predicted paintings from the test set.
- Must use elements of the features when generating the prompts
- Can we run a diffusion model locally to create the painting?

Project phases

- (1) Record 60 paintings using my cellphone for the training (hearts, waves, stars).
- (2) Check if code works (feature extraction, mapping).
- (3) Export a motion fingerprint for each painting as well as plots to explore results.
- (4) Enhance the code in a dialogue with GPT5.
- (5) Create the correctly predicted paintings from a test set of 15 paintings.

Painting with my cellphone

Using phyphox to record brush strokes

Custom Experiment Setup in phyphox

- *Created a custom phyphox experiment to record motion data*
- *Each recording corresponds to one painting*

Sensors used

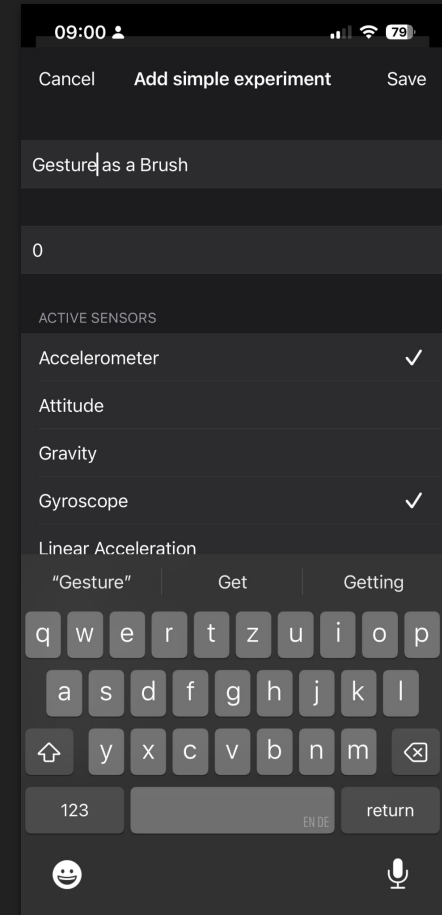
- (1) *Accelerometer (without gravity)*
- (2) *Gyroscope*

Sampling rate

- Sampling rate set to 0 (fastest possible).
- Allows to record at the max. sensor frequency (~100–200 Hz).
- Ensures fast motion and subtle gesture dynamics are preserved.

Recording process

- Each gesture set recorded for ~10 seconds:
20 painted hearts, 20 painted stars, 20 painted waves and 15 test p.
- Data exported as CSV files (e.g., Painting_1.csv).
- These recordings become the input for the AI painting pipeline.



Feature Extraction & Machine Learning

From Motion to AI prediction

Feature extraction (summary)

RMS (Root Mean Square): How strong is the movement?

- *Jerk: How sudden is the movement?*
- *Frequency: How rhythmic is the movement?*
- *Peaks: How often does the movement spike?*

Machine learning (classification)

Models used:

- *Random Forest*
- *SVM (Support Vector Machine)*
- *Logistic Regression*
- *KNN (K Nearest Neighbors)*

Requested code improvements

Additional features to improve model accuracy and cross-validation improved the accuracy from about 27% to 40%.

Number of processed paintings: 75

	painting_name	group_label	rms_accel	rms_jerk	rms_gyro	dominant_freq_hz	freq_strength	duration_s	max_jerk	accel_peak_count	gyro_accel_ratio
0	Gesture as a brush 2020-03-14 19:25:22	heart	10.730820	67.519040	1.327883	8.844653	20.162127	21.298721	288.177879	274.0	0.123754
1	Gesture as a brush 2020-03-14 19:27:32	heart	12.847481	106.299384	2.077647	1.335712	10.148964	19.456661	436.060299	240.0	0.161716
2	Gesture as a brush 2020-03-14 19:29:14	heart	15.342523	140.430685	2.630021	0.908829	10.034819	19.796252	733.475645	197.0	0.171746
3	Gesture as a brush 2020-03-14 19:29:58	heart	12.312705	180.046975	2.349379	2.365903	4.196905	17.433097	1358.258491	166.0	0.190809
	Gesture as a brush										

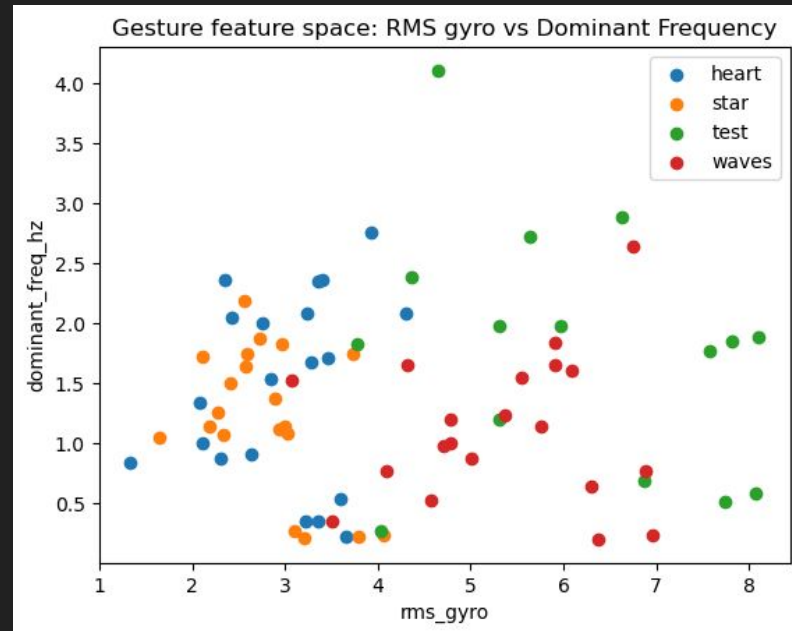
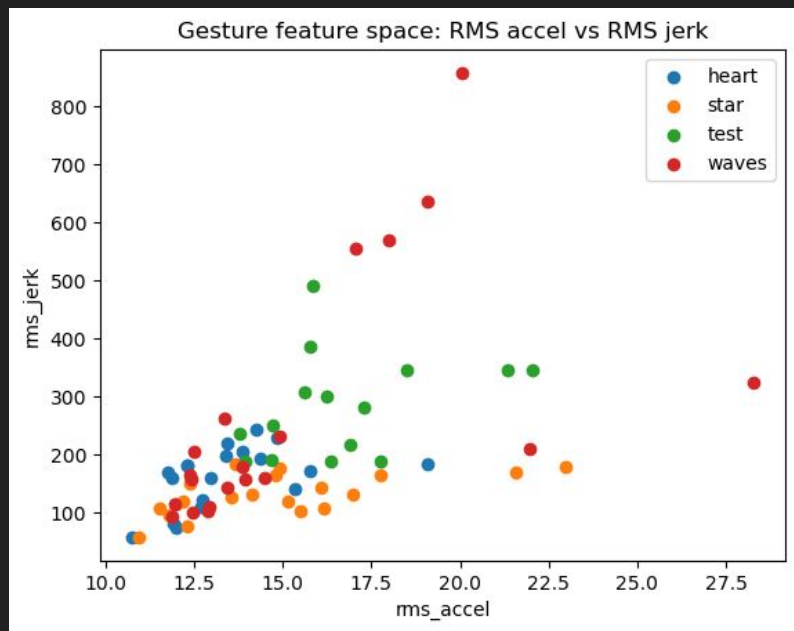
Cross-validation results (higher mean accuracy is better):

	model_name	cv_mean_accuracy	cv_std_accuracy
0	Logistic Regression	0.850000	0.062361
1	SVM (RBF kernel)	0.783333	0.066667
2	Random Forest	0.783333	0.154560
3	KNN	0.683333	0.143372

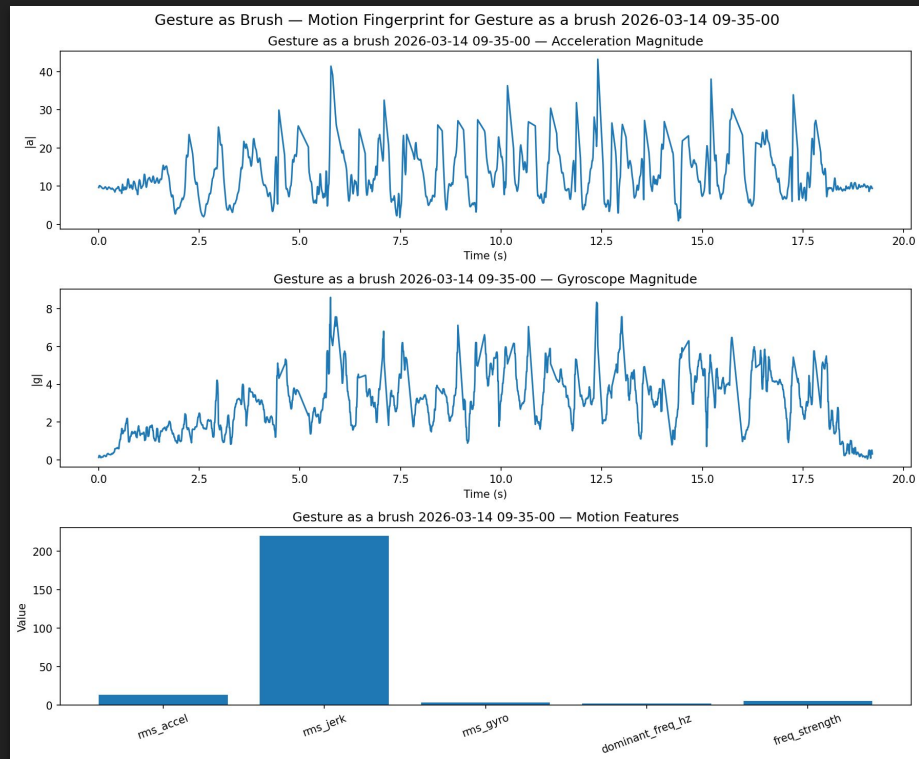
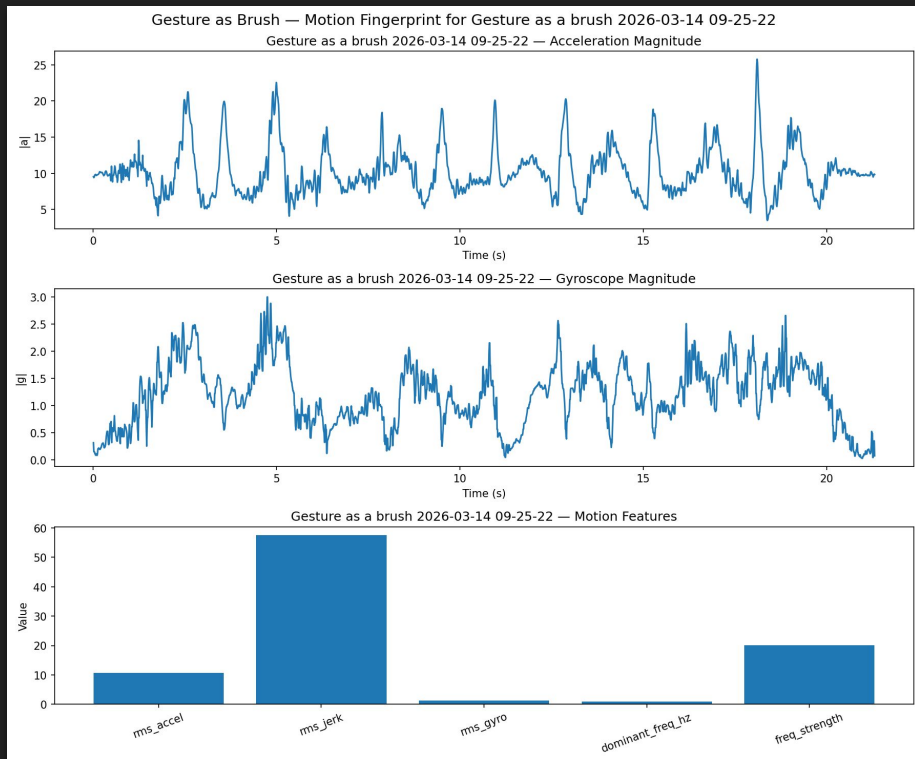
Best model selected: Logistic Regression

33											
01	8.709265	16.271519	1341.420788	180.0	0.263169						
32	4.037998	18.794641	1452.162751	161.0	0.253433						
16	15.566706	19.926356	487.337508	206.0	0.178806						
39	16.197697	20.607329	460.782056	234.0	0.192303						
50	10.008296	21.017976	1260.816079	233.0	0.180950						
11	6.656210	18.234100	2156.004216	156.0	0.250202						
33	6.543146	23.300861	1527.830370	221.0	0.238976						
14	6.161838	19.716665	1038.818138	177.0	0.184651						

Plotted Paintings



Motion Fingerprints (examples)



Prediction results & limitations

Model performance

- Accuracy: 40%
- Waves: Detected well
- Stars: Partly detected
- Hearts: Often misclassified

What the model understands:

- Movement strength (energy)
- Rhythm (repetition / frequency)

What the model doesn't understand

- Exact shape geometry
- Direction of movement in space
- Apparent issue: smooth gestures (heart, waves) look similar in the data

This would have to be further investigated!

Number of evaluated test recordings: 15
Test accuracy: 0.4

Confusion matrix:

```
[[1 0 6]
 [0 2 3]
 [0 0 3]]
```

Classification report:

	precision	recall	f1-score	support
heart	1.00	0.14	0.25	7
star	1.00	0.40	0.57	5
waves	0.25	1.00	0.40	3
accuracy			0.40	15
macro avg	0.75	0.51	0.41	15
weighted avg	0.85	0.40	0.39	15

		painting_name	true_shape	predicted_shape	correct
0	1	Gesture as a brush 2026-03-14 10-07-19	heart	waves	False
1	10	Gesture as a brush 2026-03-14 10-13-37	heart	heart	True
2	11	Gesture as a brush 2026-03-14 10-14-09	heart	waves	False
3	12	Gesture as a brush 2026-03-14 10-14-44	waves	waves	True
4	13	Gesture as a brush 2026-03-14 10-15-18	star	waves	False
5	14	Gesture as a brush 2026-03-14 10-15-53	heart	waves	False
6	15	Gesture as a brush 2026-03-14 10-17-03	heart	waves	False
7	2	Gesture as a brush 2026-03-14 10-08-58	star	star	True
8	3	Gesture as a brush 2026-03-14 10-09-51	star	star	True
9	4	Gesture as a brush 2026-03-14 10-10-26	waves	waves	True
10	5	Gesture as a brush 2026-03-14 10-10-55	heart	waves	False
11	6	Gesture as a brush 2026-03-14 10-11-30	waves	waves	True
12	7	Gesture as a brush 2026-03-14 10-11-57	heart	waves	False
13	8	Gesture as a brush 2026-03-14 10-12-30	star	waves	False
14	9	Gesture as a brush 2026-03-14 10-13-05	star	waves	False

From Prediction to (Prompted) Art

Pipeline:

gesture -> prediction -> prompt -> image

Prompts describe:

- *motion style*
- *texture*
- *energy*

Code improvement: Inclusion of specific numeric values from the feature space for more accurate representation of the movement in the generated painting

Image Generation

Specs:

- *Model: Stable Diffusion (v1.5)*
- *Runs locally on laptop*

Content:

- *Only correctly predicted gestures used*
- *Final output: ~6 paintings*

Example prompt

```
=====
PAINTING: 12 Gesture as a brush 2026-03-14 10-14-44
PREDICTED SHAPE: waves
ACTUAL SHAPE: waves
```

FEATURE VALUES:

```
rms_accel = 18.4817
rms_jerk = 345.3885
rms_gyro = 8.0776
dominant_freq_hz = 0.5874
freq_strength = 4.3048
duration_s = 16.8349
std_accel = 6.4300
max_accel = 39.6527
std_jerk = 345.3884
max_jerk = 2770.3401
std_gyro = 3.9509
max_gyro = 20.0194
accel_peak_count = 125.0000
gyro_peak_count = 83.0000
accel_zero_crossings = 94.0000
gyro_accel_ratio = 0.4371
```

PROMPT:

wave-like abstract composition, flowing repeated curves, abstract expressionist acrylic painting on canvas, visible brush strokes, heavy impasto paint, rich acrylic paint texture, moderate movement energy, confident brushwork, balanced contrast, noticeable motion accents, textured brushwork, some broken edges, few motion pulses, spacious composition, strong rotation, swirling composition and circular movement, rotation-driven gesture, spiral and circular painterly flow, weak rhythm, irregular distribution of marks, medium gesture duration, balanced coverage of the canvas, few sharp interruptions, gesture rhythm around 0.59 Hz

NEGATIVE PROMPT:

photorealistic, realistic, portrait, face, faces, people, human figure, object, objects, text, typography, letters, watermark, logo, 3d render, cartoon, anime, low resolution, blurry

```
=====
```

Resulting image (Waves)



All correctly labelled paintings (6/15): Waves



All correctly labelled paintings (6/15): Stars



All correctly labelled paintings (6/15): Hearts

